PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Reference Architecture: Civic Grievance & Complaint Management System (IBM/AWS Reference)

Date: August 7, 2026 Maximum Marks: 4 Marks

Project Name: Online Complaint Registration System (MERN Stack)

TECHNICAL ARCHITECTURE DIAGRAM OVERVIEW

USERS/CLIENTS: Citizen Web App | Mobile App (iOS/Android) | Voice/Chat Bot | ADMIN & DEPARTMENT OFFICERS
CLOUD/BACKEND PLATFORM: AWS / IBM CLOUD

Guidelines Followed:

- Included full application logic (Filing, Geo-Tagging, ML Triage, Notifications).
- Infrastructural demarcation clearly separated into Client Tier, Cloud Backend, and Admin Portal.
- External interfaces integrated: SMS/Email Notification (Twilio/SendGrid), Maps API, Identity Verification API.
- Data Storage & ML Interface indicated: Managed Relational/NoSQL databases, Object Storage, and ML Inference Models.

TABLE-1: COMPONENTS & TECHNOLOGIES

S.NO	COMPONENT	DESCRIPTION	TECHNOLOGY
1	**User Interface**	Citizen web portal and mobile application for filing complaints, tracking status, and receiving updates.	React.js, React Native, Tailwind CSS
2	**Application Logic-1**	Core Service: Complaint intake, category-based routing logic, and status workflow state machine management.	Node.js (Express) / Python (FastAPI)
3	**Application Logic-2**	AI Chatbot Assistant: Voice & Text guided complaint filing and preliminary category classification.	IBM Watson Assistant / Dialogflow
4	**Application Logic-3**	Geo-Tagging Engine: Location capture and mapping of complaints for hotspot visualization.	Google Maps API / Mapbox
5	**Database**	Primary database for structured data like user profiles, department mapping, and transaction records.	PostgreSQL / MySQL
6	**Cloud Database**	Managed NoSQL database for unstructured complaint descriptions, officer remarks, and chat logs.	MongoDB Atlas / IBM Cloudant
7	**File Storage**	Encrypted object storage for photo/video evidence and supporting documents.	AWS S3, IBM Cloud Object Storage
8	**External API-1**	SMS & Email Notification API: Filing confirmations, status-change alerts, and resolution updates.	Twilio SMS API / SendGrid
9	**External API-2**	Identity Verification API: Citizen/Officer identity verification for authenticated filings.	Aadhaar API / Twilio Verify
10	**Machine Learning Model**	Complaint Triage Model: Recommends appropriate department/category and priority based on complaint text.	Python, TensorFlow / Scikit-Learn
11	**Infrastructure**	Containerized deployment with auto-scaling to handle peak filing periods and secure API hosting.	Docker, Kubernetes (EKS / IKS), NGINX

PROJECT DESIGN PHASE-II

Online Complaint Registration System (MERN Stack) - Architectural Characteristics & References

TABLE-2: APPLICATION CHARACTERISTICS

S.NO	CHARACTERISTICS	DESCRIPTION	TECHNOLOGY
1	**Open-Source Frameworks**	Utilizes modular open-source libraries for backend service creation, frontend rendering, and asynchronous background tasks (e.g., automated email/SMS notifications).	React.js, Node.js Express, FastAPI, Celery, Redis
2	**Security Implementations**	Strict compliance with local Data Protection/RTI standards. Implements end-to-end AES-256 encryption for data at rest, TLS 1.3 for data in transit, Role-Based Access Control (RBAC), and JWT authentication.	AES-256, TLS 1.3, Auth0 / JWT, OWASP Top 10 Protections, AWS WAF
3	**Scalable Architecture**	Built using Microservices Architecture with stateless API services to horizontally scale complaint intake, geo-tagging, and user profile management independently during high-demand periods.	Kubernetes HPA (Horizontal Pod Autoscaler), Docker, NGINX Ingress
4	**Availability**	High availability designed with multi-region deployment, active-passive failover DB clusters, and cloud load balancing guaranteeing 99.9% uptime for urgent grievance filings.	AWS Elastic Load Balancing (ELB) / Cloudflare Traffic Manager, PostgreSQL Multi-AZ
5	**Performance Design**	In-memory caching for complaint-status dashboards to reduce DB queries by 80%. Content Delivery Network (CDN) for fast static asset loading and sub-100ms API response time.	Redis Cache, Cloudflare CDN, Database Indexing & Connection Pooling

PROJECT SUMMARY & WORKFLOW OVERVIEW

"Online Complaint Registration System (MERN Stack)" is a comprehensive digital governance platform designed to streamline citizen-department interactions. The system provides seamless online grievance filing, intelligent category-based triage using Machine Learning, geo-tagged evidence submission, and automated status tracking. By leveraging a scalable Microservices Architecture, the system ensures zero downtime and rapid response times even during peak filing periods. Sensitive citizen data is safeguarded using enterprise-grade encryption and role-based access management complying with data protection regulations.

REFERENCES & DOCUMENTATION LINKS

- C4 Model for Software Architecture: https://c4model.com/
- IBM Developer Architecture - Online Order & Grievance Processing: https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
- IBM Cloud Architecture Center: https://www.ibm.com/cloud/architecture
- AWS Reference Architecture Center: https://aws.amazon.com/architecture
- Guide to Technical Architecture Diagrams: https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d